AttnMPNet: An Attention-Based Message Passing Network for MIMO Detection

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***Abstract*—In this letter, we propose AttnMPNet, an attention-based message passing neural network for massive multiple-input multiple-output (MIMO) detection. AttnMPNet integrates Gaussian approximated message passing with self-attention to iteratively refine soft posterior estimates for each transmitted symbol. Within each iteration, approximate likelihood messages are first constructed and then processed through attention blocks to capture inter-symbol dependencies, followed by symbol-wise posterior probability estimation. To support this inference pipeline, we further design a compact and expressive feature representation that enables both efficient computation and high detection accuracy in massive MIMO scenarios. Simulation results show that AttnMPNet achieves near-maximum-likelihood performance across diverse MIMO configurations while significantly reducing inference latency compared to neural baselines.**

***Index Terms*—Multiple-input multiple-output, data detection, deep learning, attention mechanism, message passing.**

I. Introduction

MULTIPLE-INPUT multiple-output (MIMO) systems play a pivotal role in boosting the spectral efficiency of modern wireless networks [1]. However, reliable symbol detection in massive MIMO remains a formidable task due to severe inter-symbol interference and the exponential complexity of maximum-likelihood (ML) detection [2]. The need for efficient, high-performance MIMO detection becomes more critical under high-order modulation and large antenna arrays.

Linear detectors such as minimum mean square error (MMSE) method are widely adopted due to their low computational complexity, but they often suffer from poor detection performance, especially under strong interference [3]. Among nonlinear methods, a prominent class is based on message passing (MP), which iteratively exchanges beliefs between variable and factor nodes on the underlying factor graph [4]. Classical MP-based detectors such as approximate message passing (AMP) [5], orthogonal AMP (OAMP) [6], and expectation propagation (EP) [7] have been extensively studied for their analytical elegance. Nevertheless, their performance often deteriorates in high-loading regimes.

Recently, deep learning-based detectors have gained attention due to their superior adaptability and detection performance. Early neural detectors such as DetNet [8], OAMPNet [9], and MMNet [10] adopt a model-driven approach by unfolding iterative algorithms into neural architectures, while learning key parameters to enhance convergence speed and robustness. More recent, graph neural network (GNN)-based detectors (e.g., AMP-GNN [11] and GEPNet [12]) have been proposed to refine posterior statistics in MP inference. While effective in modeling local dependencies, these methods rely on irregular graph computations that involve numerous small kernel launches and uncoalesced memory access patterns–leading to suboptimal runtime. In a related line of work, recurrent equivariant MIMO (RE-MIMO) [13] adopts attention mechanisms to capture inter-symbol correlations. Leveraging the global modeling and parallelism of Transformer architectures, RE-MIMO offers better runtime efficiency than GNN-based counterparts. However, it redundantly injects global channel and residual information into the network, resulting in an over-parameterized model with increased training difficulty, inference latency, and limited performance. These limitations motivate a more compact, scalable, and expressive alternative that can effectively capture inter-symbol dependencies while maintaining high efficiency and accuracy.

In this letter, we propose AttnMPNet, an attention-based message passing network tailored for large-scale MIMO detection. AttnMPNet unfolds a simplified inference algorithm inspired by Gaussian AMP [4] into a neural framework, where posterior symbol features are iteratively refined via self-attention. In contrast to GNN-based detectors, our method employs highly parallelizable Transformer blocks to model inter-symbol dependencies without relying on irregular graph operations. Moreover, we design a compact and efficient feature representation that avoids redundant embeddings as seen in transformer-based detectors such as RE-MIMO [13]. This design not only reduces model complexity but also enhances feature expressiveness and improves detection performance. Extensive simulations demonstrate that AttnMPNet achieves near-ML detection accuracy with significantly faster inference compared to existing neural baselines.

The remainder of this letter is organized as follows. Section II introduces the system model and preliminaries on message passing and attention mechanisms. Section III presents the AttnMPNet architecture. Section IV analyzes complexity. Section V reports numerical results comparing

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AttnMPNet with existing detectors. Section VI concludes this letter.

II. Preliminaries

A. System Model

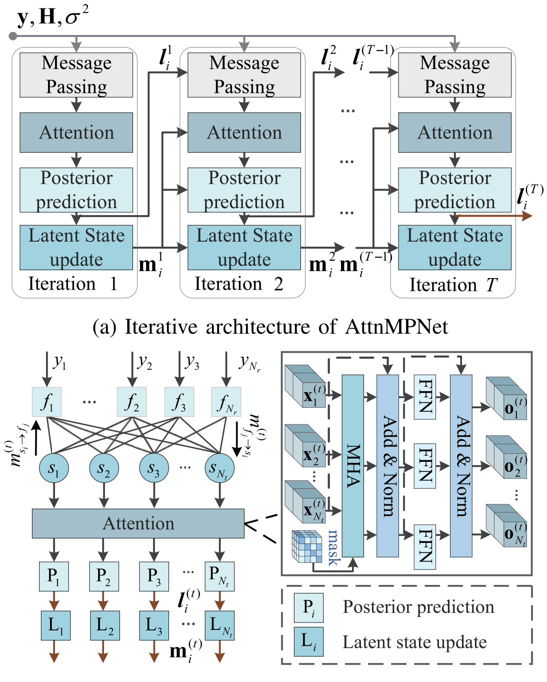
We consider the uplink detection problem in multi-user MIMO (MU-MIMO) systems, where N_t single-antenna users simultaneously transmit to a base station equipped with N_r antennas. The baseband system model can be expressed as $\mathbf{y} = \mathbf{Hs} + \mathbf{n}$, where $\mathbf{y} = [y_1, y_2, \ldots, y_{N_r}]^T \in \mathbb{C}^{N_r}$ is the received signal vector at the base station, $\mathbf{H} \in \mathbb{C}^{N_r \times N_t}$ is the channel matrix whose entries are independently drawn from zero-mean and unit-variance complex Gaussian distribution, $\mathbf{s} = [s_1, s_2, \ldots, s_{N_t}]^T \in \mathcal{S}^{N_t}$ is the transmitted symbol vector with each entry drawn independently from a discrete constellation $\mathcal{S}$. $\mathbf{n} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2\mathbf{I})$ is the additive white Gaussian noise (AWGN). The task of MIMO detection is to recover the transmitted symbol vector $\mathbf{s}$ given the received signal and known channel matrix $\mathbf{H}$. The optimal ML detection aim to find the most likely transmit vector by minimizing the Euclidean distance. However, ML detection is computationally prohibitive due to the exponential search space. Instead of computing the hard decision directly, we aim to estimate the symbol-wise posterior distributions $p(s_i \mid \mathbf{y}, \mathbf{H})$ via approximate inference. This probabilistic formulation can be naturally represented using a factor graph, where inference is performed through iterative message passing.

B. Factor Graph and Message Passing

The MIMO detection problem can be naturally structured as a fully connected bipartite factor graph, where each transmitted symbol s_i is a variable node, and each received observation y_j models a factor node f_j that encodes a local likelihood constraint. Inference on this graph involves iteratively exchanging messages between variable and factor nodes as $m_{s_i \to f_j}(s_i) \propto \prod_{k \neq j} m_{f_k \to s_i}(s_i)$ and $m_{f_j \to s_i}(s_i) \propto \sum_{\mathbf{s} \backslash s_i} \psi_j(\mathbf{s}) \prod_{l \neq i} m_{s_l \to f_j}(s_l)$, where $\psi_j(\mathbf{s}) = \exp(-\frac{1}{\sigma^2}|y_j - \sum_i h_{j,i} s_i|^2)$ encodes the local likelihood at f_j and $h_{j,i}$ denotes the (j, i)-th entry of $\mathbf{H}$. After sufficient iterations, the marginal posterior of s_i is finally approximated by $p(s_i|\mathbf{y}, \mathbf{H}) \approx \prod_{j=1}^{N_r} m_{f_j \to s_i}(s_i)$. However, exact inference on such dense and loopy MIMO graphs is computationally prohibitive due to the high-dimensional summation. This motivates the development of tractable approximate inference schemes such as AMP, OAMP, and learning-based techniques.

C. Attention Mechanism

Attention mechanisms are widely used to model pairwise dependencies among input elements [14]. In our network, we adopt the multi-head attention (MHA) architecture to enable efficient and structured feature propagation. Given an input sequence $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_L]^\top \in \mathbb{R}^{L \times d_{\text{model}}}$, MHA first projects $\mathbf{X}$ into N_h sets of queries, keys, and values using head-specific weights: $\mathbf{Q}_h = \mathbf{X}\mathbf{W}_h^Q$, $\mathbf{K}_h = \mathbf{X}\mathbf{W}_h^K$, $\mathbf{V}_h = \mathbf{X}\mathbf{W}_h^V$, where $\mathbf{W}_h^Q, \mathbf{W}_h^K, \mathbf{W}_h^V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ are learnable parameters. Here, L is the sequence length, d_{model} is the input feature dimension, and d_k is the per-head dimension. For each head h, the attention output is computed as $\text{head}_h = \text{softmax}(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}} + \mathbf{B}_h)\mathbf{V}_h$, where $\mathbf{B}_h \in \mathbb{R}^{L \times L}$ is a learnable or input-dependent attention bias that can encode prior structural information. The outputs from all heads are concatenated and linearly transformed: $\text{MHA}(\mathbf{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_{N_h})\mathbf{W}^O$, with $\mathbf{W}^O \in \mathbb{R}^{N_h d_k \times d_{\text{model}}}$ being the output projection. In practice, we set $d_k = d_{\text{model}}/N_h$ to ensure dimension consistency.



(a) Iterative architecture of AttnMPNet

(b) Intra-iteration workflow and module details

Fig. 1. The model of AttnMPNet detector.

III. Proposed AttnMPNet

In this section, we present *AttnMPNet*, a neural detection framework that integrates message passing and attention mechanisms to estimate symbol-wise posteriors in MIMO systems. As illustrated in Fig. 1, the detector is unfolded into T fixed iterations. In each iteration, we first perform Gaussian-approximated message-passing step inspired by the AMP paradigm in [4], extracting local posterior statistics for every symbol node. Then, a self-attention module models inter-symbol dependencies induced by the channel. The refined features are used to predict the posterior probabilities of each symbol, and a latent state is updated to capture temporal information across iterations. The following subsections will detail each of these modules and their interfaces.

A. Message Passing With Gaussian Approximation

At the beginning of each iteration t, the network performs a single round of message passing that yields a Gaussian belief $q^{(t)}(s_i) = \mathcal{N}_{\mathbb{C}}(s_i; \xi_i^{(t)}, \gamma_i^{(t)})$, for each symbol node s_i, which later serves as the input feature of the attention module.

We begin by applying Gaussian approximation to the standard sum-product BP algorithm on the MIMO factor graph. For each factor node f_j associated with observation y_j, we rewrite the measurement as $y_j = h_{j,i}s_i + \sum_{k \neq i} h_{j,k}s_k + n_j$. Under the Gaussian approximate inference assumption [4], [15], the interference term $\sum_{k\neq i} h_{j,k}s_k$ is approximated as a Gaussian random variable with mean $\sum_{k\neq i} h_{j,k}\hat{s}^{(t-1)}_{s_k\to f_j}$ and variance $\sum_{k\neq i} |h_{j,k}|^2\hat{\tau}^{(t-1)}_{s_k\to f_j}$, where $\hat{s}^{(t-1)}_{s_k\to f_j}$ and $\hat{\tau}^{(t-1)}_{s_k\to f_j}$ are the mean and variance contained in the symbol-to-factor message $m^{(t-1)}_{s_k\to f_j}$ of the previous iteration. This yields a Gaussian message $m^{(t)}_{f_j\to s_i}(s_i) \propto \mathcal{N}_{\mathbb{C}}(h_{j,i}s_i; z^{(t)}_{f_j\to s_i}, \nu^{(t)}_{f_j\to s_i})$, where

$$z^{(t)}_{f_j\to s_i} = y_j - \sum_{k\neq i} h_{j,k}\hat{s}^{(t-1)}_{s_k\to f_j}, \tag{1}$$

$$\nu^{(t)}_{f_j\to s_i} = \sigma^2 + \sum_{k\neq i} |h_{j,k}|^2\hat{\tau}^{(t-1)}_{s_k\to f_j}. \tag{2}$$

Assuming independence across observations, we can aggregate all messages to symbol node s_i as a Gaussian posterior $q^{(t)}(s_i) \propto \prod_{j=1}^{N_r} m^{(t)}_{f_j\to s_i}(s_i) \propto \mathcal{N}_{\mathbb{C}}(s_i; \xi^{(t)}_i, \gamma^{(t)}_i)$ with

$$\gamma^{(t)}_i = \left(\sum_{j=1}^{N_r} \frac{|h_{j,i}|^2}{\nu^{(t)}_{f_j\to s_i}}\right)^{-1}, \xi^{(t)}_i = \gamma^{(t)}_i \sum_{j=1}^{N_r} \frac{h^*_{j,i}\, z^{(t)}_{f_j\to s_i}}{\nu^{(t)}_{f_j\to s_i}}. \tag{3}$$

Conventional Gaussian BP would propagate distinct messages $m^{(t-1)}_{s_i\to f_j}$ to every factor node that is recomputed as $\mathcal{N}_{\mathbb{C}}(s_i; \hat{s}^{(t-1)}_{s_i\to f_j}, \hat{\tau}^{(t-1)}_{s_i\to f_j})$ using $m_{s_i\to f_j}(s_i) \propto \prod_{k\neq j} m_{f_k\to s_i}(s_i)$. To reduce complexity, we share one mean–variance pair $\{\hat{s}^{(t-1)}_i, \hat{\tau}^{(t-1)}_i\}$ for all j, where

$$\hat{s}^{(t-1)}_i = \mathbf{p}^{(t-1)}_i \mathbf{c}^\top,\ \hat{\tau}^{(t-1)}_i = \mathbf{p}^{(t-1)}_i \left(\mathbf{c}^2\right)^\top - \left|\hat{s}^{(t-1)}_i\right|^2. \tag{4}$$

Here, $\mathbf{c} = [c_1, c_2, \ldots, c_{|\mathcal{S}|}]$ denotes the vector of constellation points and $\mathbf{p}^{(t-1)}_i = [p^{(t-1)}_1, p^{(t-1)}_2, \ldots, p^{(t-1)}_{|\mathcal{S}|}]$ denotes the predicted symbol posterior over $\mathcal{S}$ from the previous iteration.

B. Attention-Based Feature Propagation

The above derivation relies on the assumption that observations are conditionally independent given each symbol, which may not hold in MIMO systems due to the loopy nature of the underlying factor graph. Thus, the posterior distributions of different symbols can exhibit strong statistical dependencies.

Attention-based architectures have recently been explored for MIMO detection [13] for their ability to model such inter-symbol dependencies. Building on this insight, we propose a self-attention mechanism to propagate and refine the posterior symbol features across iterations. In each iteration *t*, the feature of symbol node s_i is constructed by aggregating multiple informative components. Specifically, we include the previous latent state vector $\mathbf{m}^{(t-1)}_i \in \mathbb{R}^{d_{\text{state}}}$, the posterior statistics $\xi^{(t)}_i, \gamma^{(t)}_i$ from the message passing step, and a scalar quantity termed the normalized projected residual, defined as

$$r^{(t)}_i = \frac{\left[\mathbf{H}^H\left(\mathbf{y} - \mathbf{H}\hat{\mathbf{s}}^{(t-1)}\right)\right]_i}{\sigma^2 + \sum_{j=1}^{N_r} |h_{j,i}|^2 \cdot \hat{\tau}^{(t-1)}_i}, \tag{5}$$

which captures the symbol-wise reconstruction error projected onto the *i*-th channel direction. This scalar serves as compact yet informative feedback signal to guide iterative refinement, and due to its low dimensionality, it introduces negligible overhead to the token representation. These components are linearly embedded and normalized to form the attention input:

$$\mathbf{x}^{(t)}_i = \mathrm{LN}\left(\mathbf{W}_{\text{emb}} \cdot \left[\mathbf{m}^{(t-1)}_i; \xi^{(t)}_i; \gamma^{(t)}_i; r^{(t)}_i\right]\right) \in \mathbb{R}^{d_{\text{model}}}, \tag{6}$$

where $\mathbf{W}_{\text{emb}}$ is a learnable embedding projection matrix and $\mathrm{LN}(\cdot)$ denotes layer normalization (LN).

Following the attention formulation introduced in Section II-C, we stack symbol features $\{\mathbf{x}^{(t)}_i\}_{i=1}^{N_t}$ into a sequence $\mathbf{X}^{(t)}$ and apply a single-layer Transformer block to model inter-symbol interactions. To incorporate structural prior, we design the attention bias matrix $\mathbf{B}$ to reflect the coupling strength between transmitted symbols, based on the inner product of their corresponding channel vectors. Specifically, the (*i*, *k*)-th entry of $\mathbf{B}$, which modulates the attention bias between the *i*-th and *k*-th symbols, is calculated as

$$B_{i,k} = \alpha \cdot \frac{\left|\sum_{j=1}^{N_r} h^*_{j,i}h_{j,k}\right|}{\frac{1}{2N_t^2}\sum_{i',k'}\left|\sum_j h^*_{j,i'}h_{j,k'}\right|}, \tag{7}$$

where α is a learnable scaling factor. The matrix $\mathbf{B}$ is shared across attention heads. This structure-aware bias encourages stronger attention weights between symbol pairs with high mutual interference. By normalizing against the average channel interaction, the bias remains robust across channel realizations.

The Transformer block applies multi-head self-attention with additive bias, followed by a feed-forward network:

$$\tilde{\mathbf{X}}^{(t)} = \mathrm{LN}\left(\mathbf{X}^{(t)} + \mathrm{MHA}(\mathbf{X}^{(t)})\right) \tag{8}$$

$$\mathbf{X}^{(t)}_o = \mathrm{LN}\left(\tilde{\mathbf{X}}^{(t)} + \mathrm{FFN}(\tilde{\mathbf{X}}^{(t)})\right), \tag{9}$$

where FFN denotes a two-layer position-wise feed-forward network. The first layer expands the input from d_{model} to a hidden dimension $4d_{\text{model}}$ with GELU activation, followed by a second linear layer projecting back to d_{model}, consistent with standard Transformer configurations. Both MHA and FFN blocks are wrapped with residual connections and layer normalization. The output $\mathbf{X}^{(t)}_o = [\mathbf{o}^{(t)}_1, \ldots, \mathbf{o}^{(t)}_{N_t}]^\top$ contains the refined feature representations.

C. Posterior Prediction and State Update

After attention-based feature refinement, the network proceeds to estimate the symbol-wise posterior distributions and update the latent state for the next iteration.

1) Posterior Prediction: At iteration *t*, we denote the refined feature of symbol node s_i as $\mathbf{o}^{(t)}_i \in \mathbb{R}^{d_{\text{model}}}$, obtained from the attention module. To predict the posterior distribution over the constellation set $\mathcal{S}$, we construct an augmented predictor input by concatenating the attention-refined feature $\mathbf{o}^{(t)}_i$, the latent state vector $\mathbf{m}^{(t-1)}_i$, the residual term $r^{(t)}_i$, the posterior mean $\xi^{(t)}_i$ and variance $\gamma^{(t)}_i$ from message passing. Formally, the predictor input is

$$\mathbf{z}^{(t)}_i = \mathrm{LN}\left(\left[\mathbf{o}^{(t)}_i; \mathbf{m}^{(t-1)}_i; r^{(t)}_i; \xi^{(t)}_i; \gamma^{(t)}_i\right]\right). \tag{10}$$

The normalized vector $\mathbf{z}_i^{(t)}$ is fed into a shared multi-layer perceptron (MLP) to predict the logits $\boldsymbol{\ell}_i^{(t)} \in \mathbb{R}^{|\mathcal{S}|}$ over the constellation symbols. This predictor is implemented as a two-layer MLP with hidden dimension d_{ff}, mapping the input vector to $\mathbb{R}^{|\mathcal{S}|}$ via a GELU activation. The symbol posterior is computed by applying a softmax over the logits: $\mathbf{p}_i^{(t)} = \text{softmax}(\boldsymbol{\ell}_i^{(t)})$.

2) Latent State Update: To capture historical inference context and temporal correlations across iterations, we introduce a latent state vector $\mathbf{m}_i^{(t)} \in \mathbb{R}^{d_{\text{state}}}$ for each symbol node s_i. This latent representation serves two purposes: it aggregates long-range iterative information beyond immediate messages and enables the network to correct or refine beliefs in light of evolving residuals and posterior changes. Such a temporal memory is particularly beneficial in loopy factor graphs like MIMO, and has also been involved in related works [13].

At each iteration *t*, the latent state is updated by incorporating a rich set of features, including the attention-refined feature $\mathbf{o}_i^{(t)}$, the predicted logits $\boldsymbol{\ell}_i^{(t)}$, the posterior moments $\{\hat{s}_i^{(t)}, \hat{\tau}_i^{(t)}\}$ w.r.t. the predicted soft symbol distribution $\mathbf{p}_i^{(t)}$, and the residual change $\delta r_i^{(t)} = r_i^{(t)} - r_i^{(t-1)}$. This residual change captures the instantaneous evolution of the reconstruction error, providing dynamic cues for the state update. These features are concatenated with the previous latent state $\mathbf{m}_i^{(t-1)}$ and normalized:

$$\mathbf{u}_i^{(t)} = \text{LN}\Big(\Big[\mathbf{o}_i^{(t)};, \boldsymbol{\ell}_i^{(t)}; \hat{s}_i^{(t)}; \hat{\tau}_i^{(t)}; \delta r_i^{(t)}; \mathbf{m}_i^{(t-1)}\Big]\Big). \quad (11)$$

This representation is then fed into a two-layer MLP with hidden size $2d_{\text{model}}$ and GELU activation to produce $\mathbf{m}_i^{(t)}$.

The updated latent state $\mathbf{m}_i^{(t)}$ is used in the next iteration to construct the posterior feature input, forming a closed-loop refinement cycle. After T iterations, the final logits $\boldsymbol{\ell}_i^{(T)}$ are used to make hard or soft symbol decisions. At initialization, the latent state is set to zero, and the initial logits are obtained via MMSE estimation–an approach commonly adopted in prior MP-based detectors [9], [12].

IV. Complexity Analysis

We analyze the computational complexity of AttnMPNet. An initial MMSE step is performed with complexity $\mathcal{O}(N_t^2 N_r)$, which is amortized over T iterations. Each iteration involves Gaussian message passing ($\mathcal{O}(N_t N_r)$), self-attention ($\mathcal{O}(N_t^2 d_{\text{model}})$), and multiple lightweight MLPs-including the Transformer FFN, posterior predictor, and latent state updater-with total complexity $\mathcal{O}(N_t d_{\text{model}}^2)$. Thus, the overall per-iteration complexity is $\mathcal{O}(N_t N_r + N_t^2 d_{\text{model}} + N_t d_{\text{model}}^2)$. This is on par with RE-MIMO, which also employs attention and FFN modules. However, RE-MIMO embeds the full residual vector and channel matrix columns into every symbol token, leading to significantly larger token dimensions and heavier embedding layers. In contrast, AttnMPNet uses compact designs that avoids globally redundancy, enabling smaller embedding dimensions d_{model} without sacrificing modeling capacity. This leads to lower memory and computational cost. The advantage becomes particularly pronounced in large-scale MIMO settings, where our token dimensionality remains nearly constant as antenna size increases. In terms of wall-clock runtime, AttnMPNet is GPU-friendly due to its Transformer-style design and matrix-based operations. Detailed runtime comparisons are provided in Section V.

V. Numerical Results

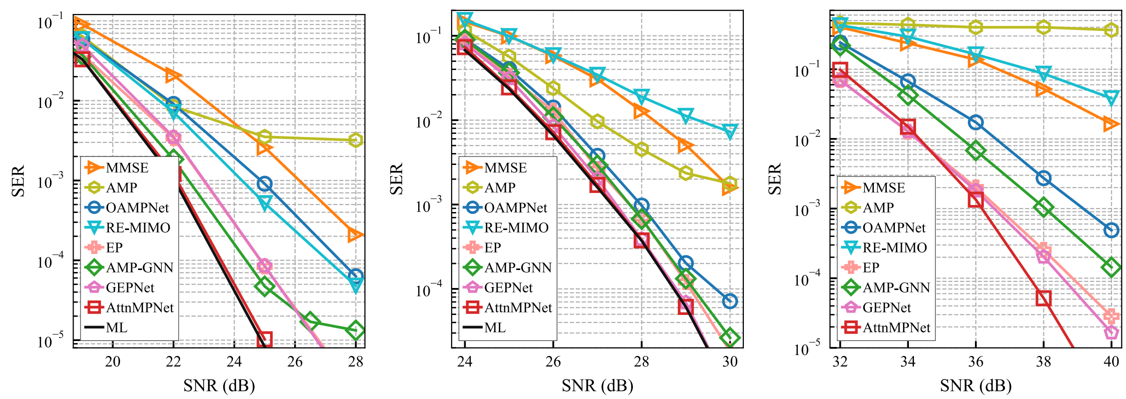
In this section, we evaluate the performance of the proposed AttnMPNet detector through comprehensive simulations under various MIMO configurations. For implementation convenience, we adopt an equivalent real-valued representation of the complex MIMO system throughout training and inference. All simulations are conducted under quadrature amplitude modulation (QAM) schemes. The signal-to-noise ratio (SNR) is defined as $\text{SNR} = \mathbb{E}[||\mathbf{Hs}||^2]/\mathbb{E}[||\mathbf{n}||^2]$.

AttnMPNet is implemented in PyTorch with fixed seeds for reproducibility. We set $T = 10$, $d_{\text{model}} = 128$, $N_h = 8$, and $d_{\text{state}} = 32$ throughout the experiments. Each epoch uses 100,000 training and 5,000 validation samples. The network is trained for 500 epochs using the Adam optimizer [16] with a 0.0001 learning rate, a batch size of 128, and a ReduceLROnPlateau scheduler. For each batch, the SNR is sampled uniformly from a range covering the waterfall region. The final models are selected based on best validation set SER. The training loss is the average cross-entropy between the predicted logits and the one-hot targets over T iterations: $\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{N_t}\sum_{i=1}^{N_t}\text{CE}(\boldsymbol{\ell}_i^{(t)}, \text{onehot}(s_i))$.

We benchmark AttnMPNet against several neural detectors, including OAMPNet [9], GEPNet [12], RE-MIMO [13], and AMP-GNN [11]. Classical baselines, MMSE, AMP [5] and EP [7], are also included for comparison. The ML bound is evaluated using sphere decoding [17], where the initial search radius is set using the ground-truth symbols to guarantee finding the exact ML solution with tractable complexity. All neural baselines are implemented using official source code. For fair comparison, iteration counts are unified to 10, consistent with the original setting in most baselines. The average symbol error rate (SER) is reported per SNR point by accumulating detection results until 1,000 symbol errors are collected.

We first examine the detection accuracy of AttnMPNet under various MIMO configurations, as illustrated in Fig. 2. In all cases, AttnMPNet consistently outperforms MMSE, AMP, as well as neural baselines including OAMPNet, RE-MIMO, AMP-GNN, and GEPNet. Specifically, compared to RE-MIMO, which also employs attention mechanisms, AttnMPNet achieves noticeable gains across all SNRs. This advantage becomes especially pronounced in large-scale and high-order modulation MIMO setups, where RE-MIMO suffers from error floors. Notably, AttnMPNet closely approaches the ML bound in Figs. 2(a) and 2(b), demonstrating both strong robustness in massive MIMO detection and effectiveness across diverse modulation orders.

To further evaluate practical efficiency, we report the wall-clock inference time of AttnMPNet and several state-of-the-art neural detectors under a 32 $\times$ 64 MIMO system with 256-QAM on an NVIDIA GeForce RTX 4070Ti SUPER GPU. Modern communication systems often demand



(a) 8×16 MIMO with 64-QAM (b) 32×64 MIMO with 256-QAM (c) 60×64 MIMO with 256-QAM

Fig. 2. SER performance comparison for different MIMO detectors.

TABLE I
INFERENCE RUNTIME (S/BATCH) OF NEURAL DETECTORS UNDER DIFFERENT BATCH SIZES. (32×64 MIMO, 256-QAM)

Batch Size	AttnMPNet	RE-MIMO [13]	GEPNet [12]	AMP-GNN [9]
1	0.0170	0.0172	0.1599	0.1458
10	0.0169	0.0208	0.1910	0.1869
50	0.0173	0.0245	0.2055	0.1936
100	0.0182	0.0458	0.3079	0.2981
500	0.0268	0.2103	1.2908	1.0730

parallel detection, mainly due to OFDM subcarrier processing and frequency-domain user scheduling. Evaluating runtime under different batch sizes thus provides practical insights into detector scalability. The superior scalability of AttnMPNet, evident in Table I, stems from its synergistic design that combines computational regularity with representation compactness. Unlike GNN-based detectors that suffer from irregular memory access patterns ill-suited for parallel hardware, AttnMPNet's operations are dominated by matrix multiplications. Notably, while RE-MIMO achieves comparable latency at small batch sizes, its runtime increases significantly with larger batches due to its heavier model width d_{model}. In contrast, AttnMPNet's fundamentally more efficient design utilizes a compact, fixed-size token representation. This results in a leaner model with a smaller memory footprint, making it inherently more scalable and better suited for high-throughput wireless systems.

VI. CONCLUSION

This letter presents AttnMPNet, an efficient attention-based message passing detector for massive MIMO systems. Built upon a simplified Gaussian inference pipeline and self-attention, the architecture enables iterative posterior refinement with high parallelism. Leveraging a compact feature representation, AttnMPNet achieves competitive detection accuracy with notably lower runtime than prior neural baselines across various MIMO configurations. Future work will explore its extension to joint detection-decoding architectures.

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